INCH-POUND
A-A-50729A
13 May 2022
SUPERSEDING
A-A-50729
15 December 1987

COMMERCIAL ITEM DESCRIPTION

METER, MULTIPLE SCALE

The General Services Administration has authorized the use of this commercial item description for all Federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID), describes a meter, multiple scale, with a rechargeable battery pack, 10 megahertz (MHz) to 18 gigahertz (GHz), which is used to measure electrical power in watts.

2. CLASSIFICATION

2.1 Classification. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall conform to class 3 performance requirements in accordance with MIL-PRF-28800. This CID makes provision for the bid sample testing in accordance with the latest MIL-PRF-28800.

2.2 Order of precedence. In the event of a conflict between the text of this CID and the references cited herein (except for associated detail specifications, specification sheets, or MS standards) the text of this CID shall take precedence. Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. SALIENT CHARACTERISTICS. The equipment shall be capable of operation within the accuracies, limits, and applicable documents.

3.1 Safety. The equipment shall be UL (Underwriters Laboratories) listed and approved and shall comply with the safety requirements of IEC 61010-1.

3.1.1 Leakage current protection. Protection shall be provided from leakage current in excess of 3.5 milliamperes (mA) voltage root mean square (Vrms) (alternating current (AC) and direct current (DC)) from any accessible conductive parts of equipment (including control shafts with knobs broken or removed and recessed calibration or adjustment controls) to either pole of the power source for any position of the power switch(es).

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: AFLCMC/EZSS, Bldg. 28, 2145 Monahan Way, Wright-Patterson AFB, OH 45433-7017 or email engineering.standards@us.af.mil. Since contact information can change, you should verify the currency of this information using the ASSIST Online database at https://assist.dla.mil.

FSC 6625

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3.1.2 Radio frequency interference (RFI) capacitors. Any capacitor used in RFI filters shall be rated for at least four (4) times the operating voltage.

3.1.3 Ventilation openings. Ventilation openings shall be designed to prevent accidental contact with live electrical circuits and mechanical moving parts.

3.1.4 Top ventilation openings. Top ventilation openings shall incorporate small mesh screening or similar design to prevent the expulsion of hot electronic particles due to failure of electronic devices such as capacitors.

3.2 Parts, materials, and processes. Except for marking, the meter shall comply with class 3 parts, materials, and process requirements of MIL-PRF-28800. Nonstandard parts approval requests shall be in accordance with the latest MIL-PRF-28800. Front panel marking shall be in accordance with MIL-PRF-28800.

3.2.1 Restricted material. The equipment shall comply with the restricted material requirements in accordance with MIL-PRF-28800, including the requirement for mercury and radioactive materials, and shall contain no combination of materials which cause deterioration of any material contained in the equipment due to the effects of outgassing.

3.3 Design and construction. Unless otherwise specified herein, the equipment shall comply with the class 3 design and construction requirements in accordance with MIL-PRF-28800. The equipment shall be of solid-state construction. Encapsulation and embedment (potting) of subassemblies may be used only for high voltage circuits where construction is susceptible to leakage or corona. All potentiometer controls shall have at least a 270 ° rotation range (4.71 radians).

3.3.1 Type of construction. Construction of the power meter shall be of a monolithic design.

3.3.2 Signal input and output connectors. Unless otherwise specified, all signal input and output connectors shall be type Bayonet Neill-Concelman (BNC) female connectors.

3.3.3 Internal cooling. The equipment shall be capable of operating continuously for twenty four (24) hours at 55 °C.

3.3.3.1 Forced air cooling motor. Direct current (DC) brushless motors shall be utilized for the forced air cooling motor in the equipment.

3.3.3.2 Air filter. Equipment utilizing a cooling scheme, which exhausts air from the equipment through the cooling fan, shall not require an externally accessible air filter.

3.3.3.3 Thermal switch. A thermal switch shall be provided which will interrupt circuit power to the meter when the internal temperature rises to the degree of possible damage to the electronic components. A minimum of 10 °C hysteresis drop from the tripping temperature must occur before circuit power is again applied.

3.4 Electrical power sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 115/230 volts AC (±10%) at line frequencies of 50 to 400 hertz (Hz).

3.4.1 Electrical power connectors. The equipment shall operate with a ground at the primary source on either side of the single phase input.

3.4.2 Power consumption. Maximum power consumption shall not exceed 20 watts.

3.4.3 Fuses. Fuse(s) shall be provided, on the exterior rear panel for replacement without entering the enclosure. The fuse(s) shall automatically disconnect the equipment from the power source in the event of excessive current requirements.

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3.4.4 Surge and transient voltage protection. The meter shall contain surge and transient voltage protection internal of the enclosure. The transient voltage protection circuit shall be so designed as to prevent electronic device damage within the system in accordance with class 3 solid state tolerance bands of MIL-PRF-28800, the surge voltage protection circuit shall conform to IEEE C62.41. The surge protection circuit shall drain excess power due to surges directly to the ground conductor for personnel protection. Normal operation of the meter may be inhibited and false operational signals may be generated under input voltage transients greater than ±20 % and frequency transient conditions greater than ±5 % from within the nominal steady-state tolerance bands of MIL-PRF-28800, damage (other than fuse failure) to the meter shall not occur during voltage transient conditions greater than ±20 % and frequency transient conditions greater than ±5 % within the maximum limits (30 % voltage transients and 10 % frequency transients) in accordance with MIL-PRF-28800. Normal operation of the meter shall be automatically resumed when the transient has passed and any blown fuses are replaced. Blown fuses shall be externally accessible. Testing shall be in accordance with MIL-PRF-28800.

3.5 Dimensions and weight. The size and weight of the equipment shall be consistent with current commercial capabilities.

3.5.1 Equipment dimensions. The nominal dimensions for the meter should be approximately 8 inches height x 8 inches width x 12 inches depth.

3.5.2 Equipment weight. The weight including all accessories (excluding manual), shall not exceed 10 pounds.

3.6 Enclosure requirements. Unless otherwise specified herein, the equipment shall comply with the enclosure requirements of MIL-PRF-28800.

3.6.1 Rackmount. Rackmount capability shall be available as an option for the equipment specified herein.

3.6.2 Tilt bail handle. A tilt-bail handle shall be provided in lieu of the front-cover storage.

3.7 Marking and identification.

3.7.1 Marking. The visible surface adjacent to panel facilities such as variable controls, indicators, jacks, connectors, and fuse holder(s), shall be marked with a suitable word, phrase, or abbreviation thereof, indicating the purpose or use of the part.

3.7.1.1 Front panel marking. The front panel marking shall comply with the panel marking requirements of MIL-PRF-28800.

3.7.2 Items to be marked.

3.7.2.1 Volume and weight. Volume and weight (including accessories) shall be marked on three external surfaces of the enclosure.

3.7.2.2 Type designation. Type designation of the AN nomenclature of the meter shall be stenciled on the exterior or mainframe on at least three adjacent surfaces, on the transient case on two opposite sides. Marking shall be in block letters approximately 2 inches in height.

3.8 Environmental requirements.

3.8.1 Temperature. Unless otherwise specified herein, the equipment shall meet its specified performance characteristics and other requirements when subjected to the class 3 temperature requirements in accordance with MIL-PRF-28800.

3.8.1.1 Non-operating temperature. The equipment shall meet its performance and accuracy requirements in a non-operating environment of -40 ° to 71 °C.

3.8.1.2 Operating temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of 0 ° to 50 °C.

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3.8.2 Humidity requirements. The equipment shall meet its required performance and accuracy requirements under humidity requirements for class 3 equipment in accordance with MIL-PRF-28800.

3.8.3 Altitude requirements. The equipment shall meet its required performance and accuracy requirements under the altitude requirements for class 3 equipment in accordance with MIL-PRF-28800.

3.8.4 Vibration requirements. The equipment shall meet the class 3 vibration requirements in accordance with MIL-PRF-28800.

3.8.5 Mechanical shock requirements.

3.8.5.1 Functional shock. The equipment shall conform to the class 3 functional shock requirement in accordance with MIL-PRF-28800.

3.8.5.2 Transit drop. The equipment shall meet transit drop requirement in accordance with MIL-PRF-28800.

3.8.5.3 Bench handling. The equipment shall conform to the specified performance and accuracy requirements in accordance with MIL-PRF-28800.

3.8.5.4 High-impact impact. The equipment shall conform to the specified performance and accuracy requirements in accordance with MIL-PRF-28800.

3.8.6 Electromagnetic interference (EMI). The equipment shall comply with the following EMI requirements in accordance with MIL-STD-461:

a. CE101: Narrow band requirements from 10 to 20 kilohertz (kHz) only.

b. CE102: Full limits.

c. CS101: Full limits.

d. CS103: Full limits.

e. RE101: Relaxed 10 decibels (dB) at the fundamental, 3rd harmonic, and 5th harmonic of the source power frequency with the measurement probe located 12 inches from device under test.

f. RE102: Full limits to 10 times the highest used or intentionally generated frequency, or 1 GHz, whichever is greater.

g. RS101: Full limits.

h. RS103: Full limits to 1 GHz.

3.8.7 Magnetic environment requirements. The equipment shall comply with the magnetic environment requirements in accordance with MIL-PRF-28800.

3.9 Reliability. Reliability verification testing, shall be in accordance with the class 3 reliability requirements in accordance with MIL-PRF-28800.

3.9.1 Mean time between failures (MTBF). The equipment MTBF shall be 2,000 hours.

3.9.2 Pre-condition requirement. Each deliverable equipment shall be subjected to the environmental screening effects of burn-in and thermal cycling, with thermal cycling being also accomplished in the sample testing phase.

3.9.2.1 Burn-in. Burn-in shall expose the equipment to extended periods of elevated temperatures, (43 °C for a minimum of 96 hours). Each equipment which fails during burn-in shall be repaired and fully retested, until failure free. Failures which occur during the burn-in test shall be noted and reported, but shall not count toward the establishment of MTBF. Prior to and following burn-in, a satisfactory operation test (SOT) shall be performed. A satisfactory operation check (SOC) shall be performed conducted at selected times during the burn-in test.

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3.9.2.2 Temperature test procedures. Temperature test procedures for class 3 equipment shall expose the equipment to hot and cold cycles in accordance with MIL-PRF-28800. Each equipment which fails during the temperature test shall be repaired and a full hot and cold cycling test shall be noted and reported, but shall not count toward cycling. The SOC shall be conducted during the last two upper and lower temperature limits.

3.10 Calibration and maintenance adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules.

3.10.1 Maintainability. The equipment shall meet the maintainability requirements in accordance with MIL-PRF-28800.

3.11 Interchangeability. The equipment shall meet the interchangeability requirements in accordance with MIL-PRF-28800.

3.12 Performance characteristics. The power meter shall come with a self-balancing analog meter, a rechargeable battery pack, and detector in kit form.

3.12.1 Frequency range. The meter and detector(s) shall have a minimum frequency range of 10 MHz to 18 GHz.

3.12.2 Detector(s). The meter and detector(s) shall come with an N type connector and a minimum of four (4) foot interface cable.

3.12.3 Front panel scale. The front-panel scale shall read in decibel-milliwatts (dBm) or watts front-panel selectable.

3.12.4 Scale range. The scale power range shall be a minimum of 10 microwatts to 10 milliwatt front-panel range selectable. The scale dBm shall be a front-panel selectable minimum range of -20 dBm to +10 dBm.

3.12.5 Calibration factor control. The meter(s) shall come with a front-panel calibration factor control.

3.12.6 Zero control. The meter shall come with a front-panel zero control.

3.12.7 DC output. The meter shall have a minimum of one (1) volt DC output proportional to power.

3.12.8 Battery pack. The meter shall contain a rechargeable battery pack. The battery pack shall provide a minimum of five (5) continuous operational power.

3.13 Manuals. The meter shall come with two (2) sets of operating and instructional manuals.

3.14 Calibration interval. After calibration, the equipment shall meet each performance requirement within the tolerances specified for a period of at least 9 months, based on an operating time of 1500 hours.

4. REGULATORY REQUIREMENTS

4.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered, recycled, environmentally preferable, or biobased content to the maximum extent practicable in accordance with Federal Acquisition Regulation (FAR) paragraphs 23.403 and 23.404(b).

4.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR paragraph 23.404(b) “Exemptions” applies and states the (1) Agency affirmative procurement

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programs must require 100 % of purchases of the Environmental Protection Agency (EPA) or the United Stated Department of Agriculture (USDA) designated items contain recovered material or biobased content, respectively, unless the item cannot be acquired: (i) Competitively within a reasonable timeframe; (ii) Meeting reasonable performance standards; or (iii) Subcontractors comply with this requirement.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall conform to the producer's own drawings, specifications, standards, and quality assurance practices and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5.2 Certification. The contractor shall certify that the product offered meets the salient characteristics of this description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace. Small businesses that cannot offer products with established commercial market acceptability, may compete by submission of bid samples.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Government documents.

7.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks listed below form a part of this specification to the extent specified herein.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-28800	General Specification for Test Equipment for use with Electrical and Electronic Equipment

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-461	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment

(Copies of these documents are available at https://assist.dla.mil.)

7.1.2 FAR and CFR. Electronic copies of the FAR may be obtained from https://www.acquisition.gov. Electronic copies of the CFR are available at https://www.ecfr.gov/cgi-bin/ECFR?page=browse.)

7.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 61010-1	Safety Requirements for Electrical Equipment for Measurement, Control, and Laboratory Use – Part 1: General Requirements

(Copies of these documents are available online at https://www.iec.ch.)

INTERNATIONAL OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C62.41	IEEE Recommended Practice on Surge Voltages in Low-Voltage AC Power Circuits

(Copies of these documents are available online at https://www.ieee.org.)

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7.3 National Stock Number (NSN). The following product(s) is the NSN assigned that corresponds to this CID. This list may not be indicative of all possible NSNs associated with the CID.

MANUFACTURER	PART NO.	NSN
Hewlett-Packard, Wattmeter	432A	6625-00-436-4883
Wavetek, Radio Frequency Meter	1034A	6625-01-080-7690
Radio Frequency Power Test Set	42B-41-4E	6625-01-069-4970

7.4 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.

b. Product conformance provisions.

c. Packaging requirements.

7.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

MILITARY INTERESTS:

Custodian:
Air Force – 184
Navy – EC
DLA - CC

Preparing activity:
Air Force – 184

Review activity:
Army – CR

Agent:
Air Force – 110

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